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Animal Shelter Capstone Project

DATA 205

CRN: 33155

Introduction:

My capstone project plan will focus on animal welfare, primarily in Montgomery County. The two datasets are from Data Montgomery and are named OAS - Animal Impound and OAS – Animal Shelter Pathway**.** The two datasets contain information on animals impounded by the county shelter. The shelter is an open-admission shelter that takes in lost, abandoned, and surrendered pets, as well as injured/ill wildlife.

Data:

The “Animal Shelter Pathway” dataset is rich in animal characteristics such as Animal type, Breed, Sex, and Age. The “Animal Impound Dataset” contains more key information, such as intake type, outcome, time of entry/exit, and shelter location, including longitude/latitude data. The primary objective is to analyze these variables over a one-year period to provide the most current trends and findings.

Goals:

The goal of the capstone project is to analyze shelter data by examining intake type, breed, age, length of stay, outcome, and other variables to determine which factors have the greatest influence on animal outcomes. Providing valuable information to pet owners surrendering their pets and shelters to reduce impounds and improve adoption rates. This topic is very important because many of the animals enter as strays or are surrendered by their owner for many reasons. By examining variables such as breed, animal type, sex, and length of stay, the project will identify patterns that shed light on what might be expected of these animals entering the system and what outcomes they face. This information is especially important for shelters and the community to improve the chances of the animal finding a new home.

Tools, methods and resources:

I will be using R for the first part of my project due to its statistical analysis and easy data visualization. The methods that I will use in R are summary statistics like averages, grouping, and counts. The main library I will use is ggplot2, along with the already installed R libraries dplyr and tidyr. The other program I will be using is Tableau to create the GIS map for my project and a dashboard. I find this program to be very user-friendly and much easier to create a GIS map than R.

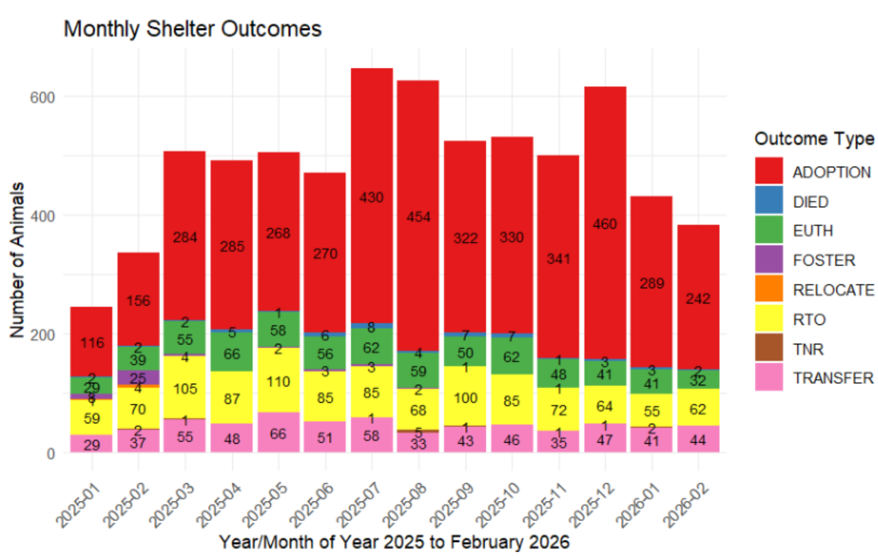
Data cleaning/pre-processing:

The two datasets were joined easily with a left join, as both datasets had the same variable, Animal ID. Through the join, I was able to keep around 85% of each dataset in the combined dataframe for the whole project. The process for the cleaning was fairly simple, involving renaming columns, which required the removal of periods and underscores. This made it easier to recall variables in my R code and in the creation of the graphs. The other issue was creating a new time variable to calculate the total time each animal spent from when they arrived until their outcome. This required the arrival and departure date to be set to a POSIT time variable, and using the difftime function to get the difference between the times in days.

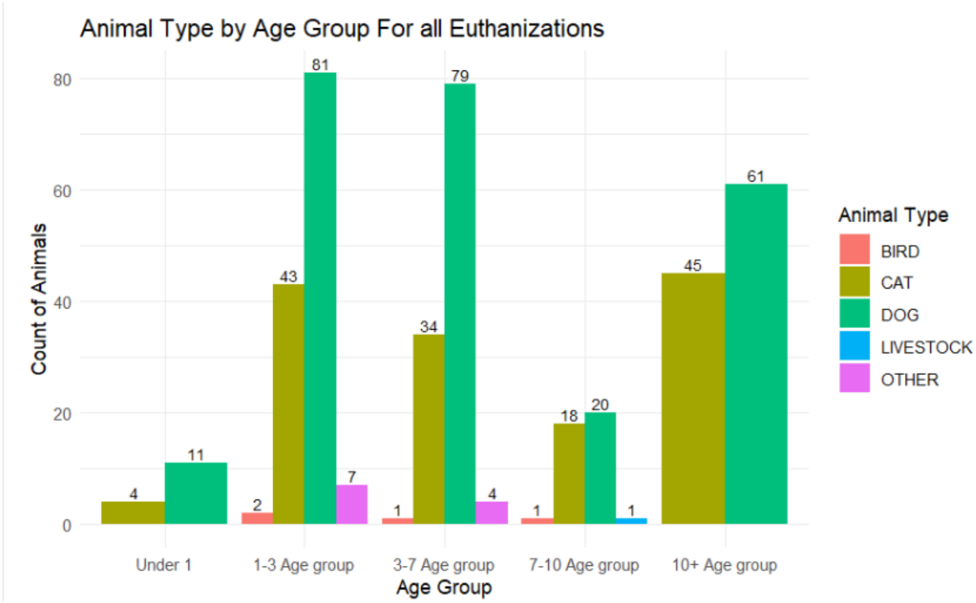
Basic descriptive statistics:

The combined Dataset has a lot of good variables that I was able to explore. The key variables I examined were the Age, Bite history, Breed, Outcome, and Animal type. Another variable I used was the days-spent variable for each animal, which I calculated from the arrival and departure dates. The two quantitative variables I had a mean analysis on were the age and the days spent. The mean for days spent was 17.30, while the age mean was 3.74. The age mean surprised me the most because I thought it would be at least 5+, but many animals are entering the shelter system at a very young age. The number of days spent is also concerning because 17 days is a long time for these animals to be without homes, and this is what I proposed to help reduce the time by examining data. Through a simple percentage table, it showed that most of the animals are entering the shelter by being strays and owners surrendering. The strays being 36.2%, while the owner surrender being 29.7%.

Description of your final data product:

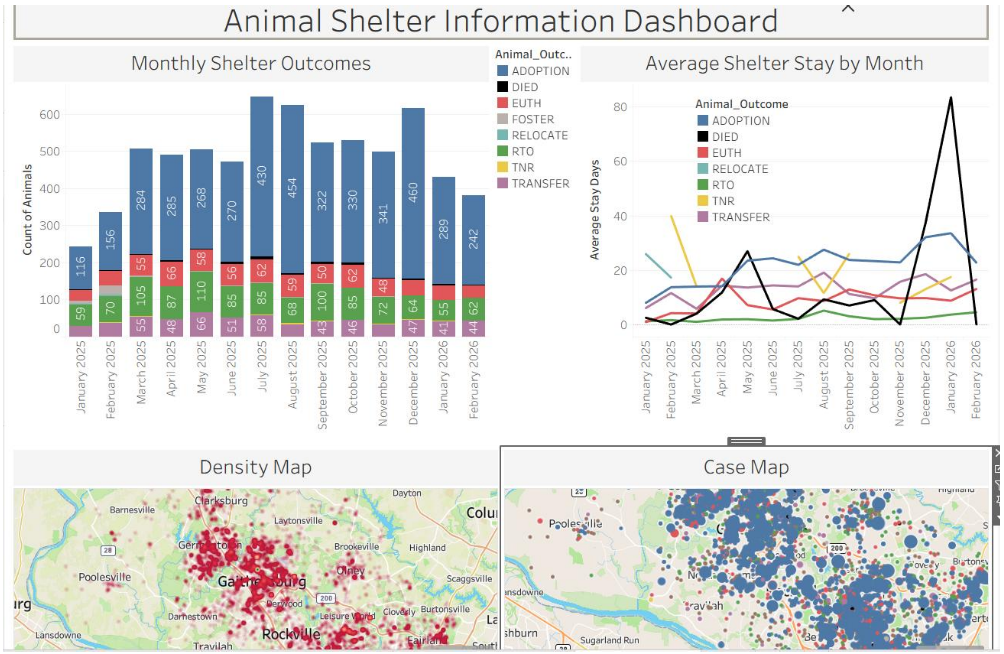


The bar chart shows that throughout July, August, and December, these are the months that have the most impounds. The big observation is that Adoption ranks the highest outcome for each month for the animals. This is a huge positive for the animals, and the second ranking outcome is the return to the owner, which is seen in the yellow. This shows that the shelter system does a good job in returning lost animals to their owners. However, the big issue is that the numbers for euthanizing are high, and this is where my project will focus on reducing by analyzing the trends of the animals that fall under this terrible outcome.

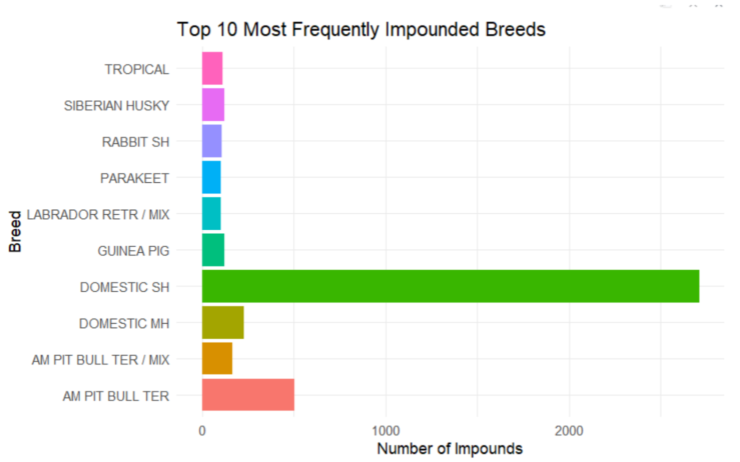


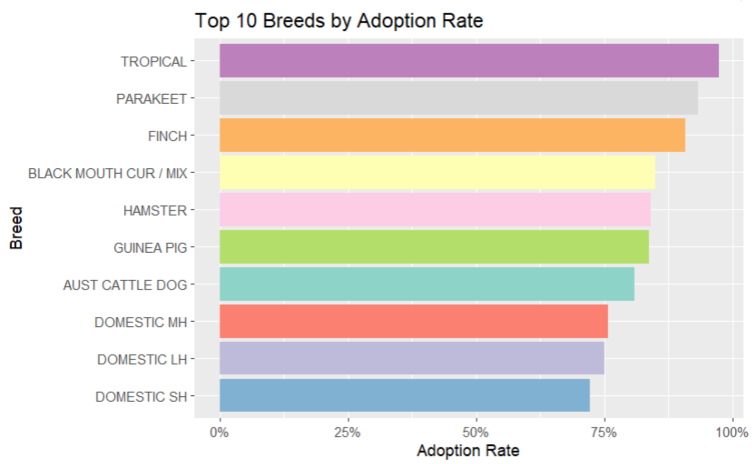
This bar chart clearly shows the age distribution of the animals that were euthanized. Most of the animals euthanized fall in the ages of 1- 7, and mostly dogs. However, the 10+ age group only consists of dogs and cats, and is the third largest age group for euthanizing. The main observation is that dogs and cats make up most of the groups throughout every age group, but most animals are euthanized in the ages 1-7 and 10+.

The line chart shows mostly an upward trend peaking in August and December. In August, the average animal stayed around 22 days, while in December it peaked at 26 days. These two numbers are way too long, and it's a sad analysis that around these months are also when there are the most animals, as the shelters try to keep the animals as long as possible so they can be able to get adopted. Numbers around the start of the year of 2025 should be the average throughout the year of <10 days on average. But the averages kept going up through the year and started to fall in February.



This is a screenshot of the Tableau Dashboard, which shows the adaptation of the graph previously split into different line graphs based on the outcome. When the user clicks on any of the outcome line graphs, it will filter that month on the case map so the user can see all the cases for that specific outcome in that month. A cool feature is that if you click on the month in the line graph, it will filter every outcome for that month on the case map. This gives the user the ability to explore where these animals are coming from. There is also a density map to show the hotspots of where most animals are coming from.





The top graphs show that most of the animals being impounded are the Domestic Shorthair Cat being over 2,500 cases. Following the shorthair will be the American Pitbull Terrier with around 500 cases. These two animals are the most impounded, but only the Domestic Shorthair pops up in the top 10 most adopted breeds. It goes to show that the reputation the Pitbull has terrible toll for being adopted in Maryland. Sadly, in Prince George, they aren't allowed to be adopted, so shelters are forced to look for rescue groups outside the county (Leybengrub, 2025). However, Shelter is striving to reach 40 dog adoptions or foster a week to avoid overcrowding (Leybengrub, 2025). These two graphs highlight that even the most impounded animals aren't going to be the most adopted in percentage, as more niche pets like tropical, Parakeet, and Finches. This suggests that owners who are looking to surrender their Pitbulls or Shorthair to do everything possible to keep the pet or give it to a family member.

Conclusion:

Overall, the capstone project highlights that animal outcomes are influenced by factors such as age, breed, and length of stay. While adoption stays the ranking outcome throughout the year, euthanizations are trailing behind adoptions. While the return to owner outcome is trailing behind adoptions, showing that the shelter system is returning animals to their owners and giving the majority of the animals a second chance in a new home. Euthanasia is still a problem that affects different age groups, intake type, animal type, and average stay time of the animals. Through age groups, the animals that are in the 3-7 age range are euthanized at a higher rate than their younger and older counterparts. Dogs are also the number one animal euthanized across all age groups, and cats are trailing behind them. The data also shows that when the average stay of all euthanized animals is over the time of all adopted animals, this is when the euthanized outcome peaked. The longest peak times of all outcomes happened in the months of July, August, and December, leading to high volume and overcapacity at the shelter. For December, this spike can be explained by the recent cutback done by President Trump that caused a lot of people to lose their federal jobs, which may have resulted to many owners deciding to surrender their pets to make ends meet (Leybengrub, 2025). The most impounds being short-haired cats and Pitbulls. Pitbulls being banned in counties like Prince George and pet insurance having a higher premium due to their bad image means that owners have to surrender or abandon their Pitbulls (Leybengrub, 2025). However, as explained by Maria Anselmo, cats, kittens, and smaller dogs are adopted quickly (Leybengrub, 2025). As shown in the most adopted breeds graph, three different cat breeds are in the top 10 breeds, such as the shorthair. Unlike the Pitbull, it doesn't rank among the top 10, even with high impounds. These patterns suggest that both seasonal effects and breed-related challenges play a major role in shelter outcomes. The datasets have provided special insight into what factors have a great influence on the outcome of the animals. Looking at graphs that highlight how age groups, breed, location, stay, and outcome distribution can help shelters better manage and prepare for the overcrowding. To support more programs like animal food drives in areas with a lot of owner surrenders. Even for owners and future owners, see which animals need the most help to either foster or adopt. This is a community and shelter issue, and both need work simultaneously to help the animals as they get stressed and in fear being so long at the shelters. They have no voice, and through this project, I hope that I touched your heart to adopt or foster, as all these animals only know how to love.

Acknowledgements:

I would like to acknowledge my professor, Lori Perine, for all the help and guidance she provided to get the resources and advice to better my project. I will also acknowledge Victoria Liu for helping me find people who are in charge of the two datasets for further questions on the variables and questions about my results. Thank you to Maria Anselmo for answering all the questions I had on the datasets.

References and Resources

Leybengrub, N. (2025, October 6). *Economic hardship drives overcrowding at Maryland's animal shelters*. The Banner. https://www.thebanner.com/community/local-news/housing-insecurity-animal-shelter-overcrowding-NDIV52NSC5DRHKDBUAUYF26HCM/

Maryland demographics: Maryland Business Data. business.maryland.gov. (2025, January 13). https://business.maryland.gov/plan-your-move/demographics/

Montgomery County, M. (2026, May 10). *Oas - animal impound: Open Data Portal*. OAS - Animal Impound | Open Data Portal. https://data.montgomerycountymd.gov/Public-Safety/OAS-Animal-Impound/6nf9-ewgt/about_data

Montgomery County, M. (2026b, May 10). *Oas - Animal Shelter Pathway: Open data portal*. OAS - Animal Shelter Pathway | Open Data Portal. https://data.montgomerycountymd.gov/Public-Safety/OAS-Animal-Shelter-Pathway/hsz7-ef2y/about_data